

Nonnegative and positive factorizations

Rational factors for rational completely positive boundary matrices

Difficulty: extreme **Importance:** interesting to the community

Status: open; checked 2026-09-08

Area: exact certificates for nonnegative symmetric factorization

Context and notation

A symmetric matrix A is **completely positive** if $A = BB^T$ for an entrywise nonnegative real matrix B with finitely many columns. Let $\mathcal{CP}_n$ be the cone of these matrices of order n . Its **cp-rank**, $\text{cpr}(A)$, is the minimum number of columns in such a factor, with $\text{cpr}(0) = 0$. Boundaries and interiors use the usual Euclidean topology on the vector space of real symmetric matrices.

Problem statement

For every integer $n \geq 5$ and every

$$A \in \mathbb{Q}^{n \times n} \cap \partial \mathcal{CP}_n,$$

does there exist a finite integer $m \geq 1$ and a matrix $B \in \mathbb{Q}_{\geq 0}^{n \times m}$ such that

$$A = BB^T?$$

The width m is unrestricted. In particular, it need not equal the real cp-rank or the order of A . This is an existence question for an exact rational certificate of complete positivity.

Rational matrices in the interior of $\mathcal{CP}_n$ have rational completely positive factors. Several boundary classes are also known, including matrices of rank at most two. The question concerns the remaining boundary matrices.

References

Abraham Berman and Naomi Shaked-Monderer, *Completely Positive Matrices over Sets*, Communications in Optimization Theory (2025), article 23, §2, Question 2.1 and the explicit boundary Open Problem, p. 2. Mathieu Dutour Sikirić, Achill Schürmann, and Frank Vallentin, *Rational factorizations of completely positive matrices*, Linear Algebra and its Applications **523** (2017), 46–51, Theorem 1.1. Max Pfeffer and José Alejandro Samper, *The Cone of 5×5 Completely Positive Matrices*, Discrete & Computational Geometry **71** (2024), 442–466, §6, Problem 6.3, discusses the order-five boundary.

Status evidence

Searches for “rational completely positive 2026 boundary” and “completely positive rational 2025” located no general boundary resolution. Alexander Oertel and Achill Schürmann’s *Generalized Perfect Matrices*, latest arXiv v2, submitted 2026-06-04, §6.1.1, still guarantees rational factors for interior matrices; its introduction describes the general rational-factorization algorithm conditionally on existence. Neither result resolves the displayed boundary question.